

From Fence Bounds to the Cubic Formula

A Self-Contained Derivation Without Cardano's Method

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Abstract. We derive the cubic formula from scratch using a method that is distinct from Cardano's classical approach. Starting from the elementary observation that $x^2 \geq 0$ for all real x , we establish fence bounds — closed intervals guaranteed to contain all real roots of a polynomial, determined by coefficient ratios alone. For the quadratic, the fence directly yields the quadratic formula via completing the square. For the depressed cubic $u^3 + pu + q = 0$ with $p > 0$ (one real root), the fence identifies the decomposition of the cubic as a pure cube plus its inflection tangent. A calculus-motivated substitution $u = t - p/(3t)$, uniquely determined by the requirement to absorb the linear term into the cube, reduces the cubic to a quadratic equation in t^3 , which is solved by the degree-2 result. Each degree's fence result feeds the next: the quadratic formula (from the degree-2 fence) solves the resolvent of the cubic (from the degree-3 fence). No step invokes Cardano's $s + t$ decomposition.

Part I: The Fence Bound

1. The Core Observation

For any real number x , the square x^2 is non-negative. If r is a real root of a polynomial, then $r^2 \geq 0$. This elementary fact, applied to the polynomial equation evaluated at a root, produces directional inequalities that constrain where real roots can lie.

2. The Quadratic Fence

Consider $x^2 + Bx + C = 0$. At any real root r :

$$r^2 = -Br - C \geq 0$$

Therefore: $Br \leq -C$, giving:

$$\text{If } B > 0: \quad r \leq -C/B \quad | \quad \text{If } B < 0: \quad r \geq -C/B$$

Theorem 1 (Quadratic Fence Bound). All real roots of $x^2 + Bx + C = 0$ lie on the same side of the fence point $f = -C/B$. When $B > 0$, both roots satisfy $r \leq f$. When $B < 0$, both satisfy $r \geq f$.

Proof. At a real root r , $r^2 = -Br - C \geq 0$ gives $Br + C \leq 0$. For $B > 0$: $r \leq -C/B$. For $B < 0$: $r \geq -C/B$. Since this holds for every real root, all roots lie on the same side. ■

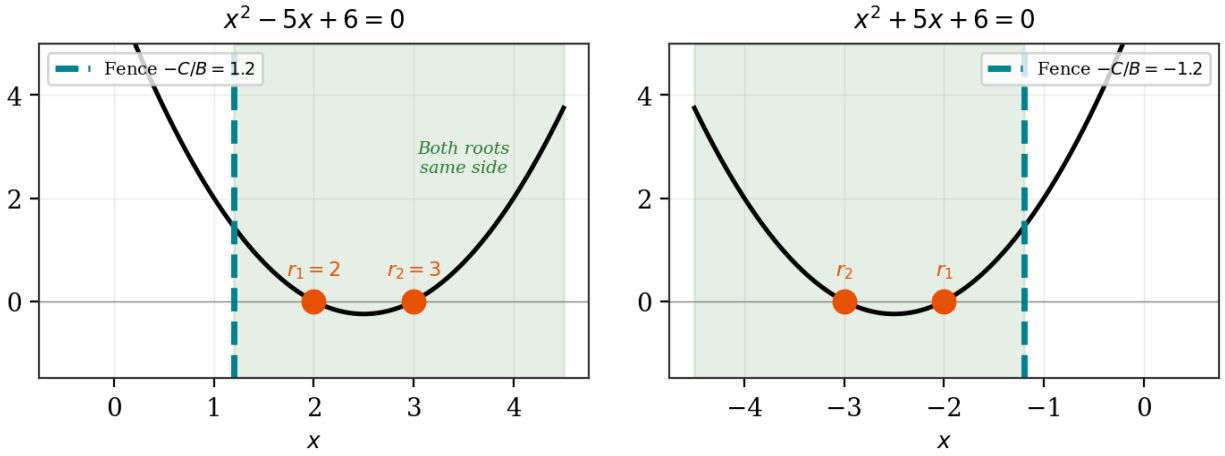


Figure 1. Quadratic fence bound. Left: $x^2 - 5x + 6 = 0$ ($B < 0$, fence at $-C/B = 1.2$, roots 2, 3 to the right). Right: $x^2 + 5x + 6 = 0$ ($B > 0$, fence at -1.2 , roots -3, -2 to the left). Green shading: allowed region.

3. The Cubic Fence (Direction 1)

Consider the depressed cubic $u^3 + pu + q = 0$ with coefficients $A = 1$, $B = 0$, $C = p$, $D = q$. At any real root r , multiply through by the non-negative quantity r^2 :

$$r^2(r^3 + pr + q) = 0$$

Since r is a root, this is satisfied identically. Instead, use the polynomial equation to write $r^3 = -pr - q$, and observe that $r^2 \geq 0$ implies $r^2 = r \cdot r \geq 0$. The constraint arises from the sub-polynomial: at a root, $r(Ar^2 + Br + C) = -D$, and since $Ar^2 + Br + C$ is a quadratic in r with known coefficients, its sign at the root constrains r .

For the depressed cubic with $A = 1$, $B = 0$, $C = p$, $D = q$: the fence points are $\alpha = -B/A = 0$ and $\beta = -D/C = -q/p$.

Theorem 2 (Cubic Fence Bound). For the depressed cubic $u^3 + pu + q = 0$ with $p > 0$ (one real root), the unique real root lies in the closed interval $[\min(0, -q/p), \max(0, -q/p)]$. The function values $f(0) = q$ and $f(-q/p) = -q^3/p^3$ have opposite signs, confirming the root is trapped between the fences by the Intermediate Value Theorem.

Proof. With $p > 0$, $AC = 1 \cdot p > 0$, so the bounded case applies: the real root lies between the fence points 0 and $-q/p$. At these points: $f(0) = q$ and $f(-q/p) = (-q/p)^3 + p(-q/p) + q = -q^3/p^3 - q + q = -q^3/p^3$. Since $p > 0$, these have opposite signs regardless of $\text{sgn}(q)$, so the IVT guarantees a root strictly between them. ■

One-real-root detection. For a depressed cubic $u^3 + pu + q = 0$, the condition $p > 0$ guarantees exactly one real root. This is because $f'(u) = 3u^2 + p > 0$ for all u when $p > 0$, making f strictly monotone increasing. A strictly monotone continuous function crosses zero exactly once. No discriminant computation is needed.

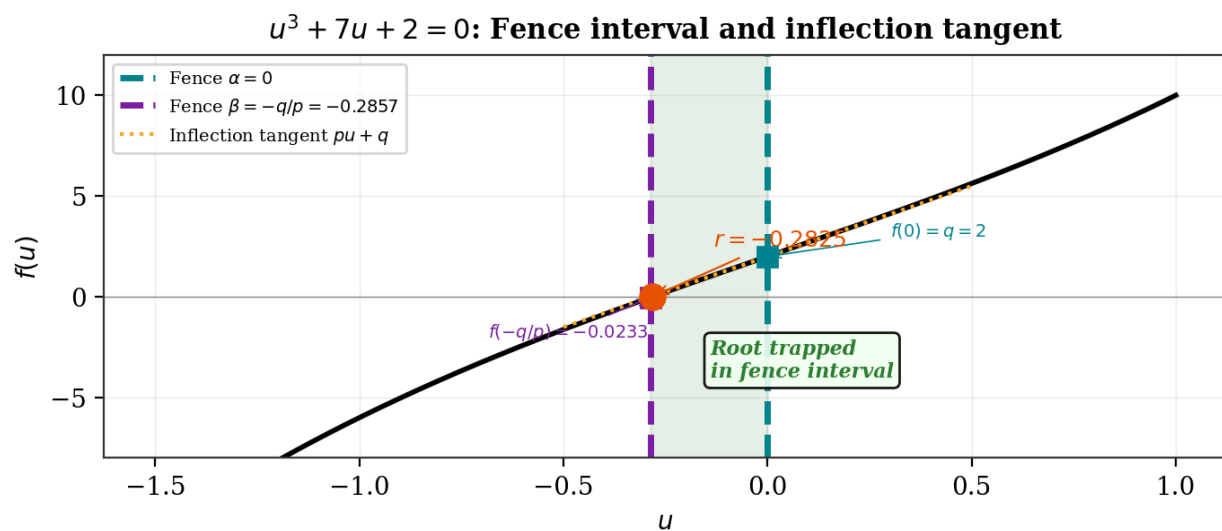


Figure 2. Cubic fence bound for $u^3 + 7u + 2 = 0$. Fence points at $\alpha = 0$ (cyan) and $\beta = -q/p = -2/7$ (magenta) bracket the root (orange). Gold dashed line: the inflection tangent $pu + q$.

Part II: From Fence to Formula

4. Degree 2: Fence → Quadratic Formula

The quadratic $x^2 + Bx + C = 0$ can be solved by completing the square — the standard technique. Here we show how the fence normalization leads to the same result.

Normalize by the fence point: set $x = (-C/B) \cdot g$, and define $\mu = C/B^2$. Substituting:

$$\mu g^2 - g + 1 = 0$$

This is a quadratic in g . Complete the square:

$$(g - 1/(2\mu))^2 = (1 - 4\mu)/(4\mu^2)$$

$$g = (1 \pm \sqrt{1 - 4\mu}) / (2\mu)$$

Substituting back $x = (-C/B) \cdot g$ with $\mu = C/B^2$:

Theorem 3 (Quadratic Formula from Fence). The roots of $x^2 + Bx + C = 0$ are:

$$x = (-B \pm \sqrt{B^2 - 4C}) / 2$$

The discriminant $B^2 - 4C$ emerges from the fence normalization parameter $\mu = C/B^2$: real roots exist when $\mu < 1/4$, equivalently $B^2 - 4C > 0$.

5. Degree 3: Fence → Inflection Tangent → Cubic Formula

5.1 The Inflection Tangent Decomposition

The depressed cubic $f(u) = u^3 + pu + q$ has its inflection point at $u = 0$ (where $f'(0) = 3 \cdot 0 = 0$). The tangent line at the inflection point is:

$$L(u) = f'(0) \cdot u + f(0) = pu + q$$

This is the fence's key structural insight: the depressed cubic is a pure cube plus its own inflection tangent:

$$f(u) = u^3 + (pu + q)$$

Finding the root means solving $u^3 = -(pu + q)$. The cube u^3 must exactly cancel the inflection tangent at the root.

5.2 The Absorption Question

This decomposition motivates a precise question from calculus: is there a substitution $u = h(t)$ such that $u^3 + pu$ becomes a pure function of t , with no linear or reciprocal terms? Try the natural two-term ansatz:

$$u = t + k/t$$

for a constant k to be determined. Expanding u^3 :

$$u^3 = t^3 + 3kt + 3k^2/t + k^3/t^3$$

Adding $pu = pt + pk/t$:

$$u^3 + pu = t^3 + k^3/t^3 + (3k + p)t + (3k^2 + pk)/t$$

To eliminate both the t and $1/t$ terms, set $3k + p = 0$:

$$k = -p/3$$

Check: the $1/t$ coefficient becomes $3(-p/3)^2 + p(-p/3) = p^2/3 - p^2/3 = 0$. Both terms vanish simultaneously.

Theorem 4 (Absorption Substitution). The substitution $u = t - p/(3t)$, uniquely determined by the requirement that $u^3 + pu$ contains no terms in t or $1/t$, gives:

$$u^3 + pu = t^3 - p^3/(27t^3)$$

The linear term of the inflection tangent has been completely absorbed into the cubic structure.

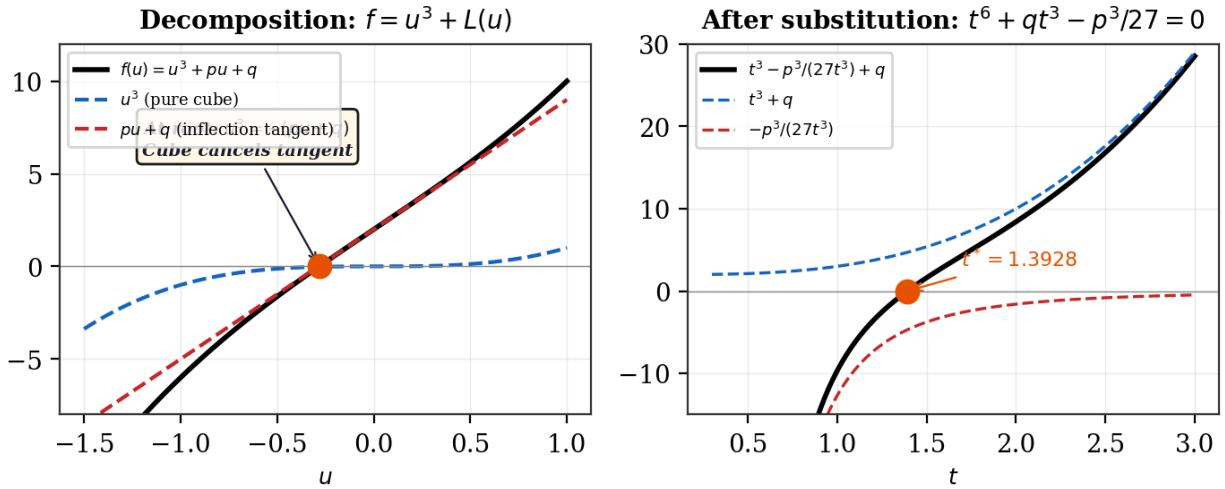


Figure 3. Left: the depressed cubic decomposed as u^3 (blue) plus inflection tangent (red). At the root, the cube cancels the tangent. Right: after substitution, the equation in t has no linear terms — it is a quadratic in t^3 .

5.3 Reduction to a Quadratic

Substituting into $u^3 + pu + q = 0$:

$$t^3 - p^3/(27t^3) + q = 0$$

Multiply through by t^3 :

$$t^6 + q \cdot t^3 - p^3/27 = 0$$

Theorem 5 (Reduction to Degree 2). Setting $w = t^3$, the depressed cubic $u^3 + pu + q = 0$ reduces to the quadratic:

$$w^2 + qw - p^3/27 = 0$$

which is solved by the degree-2 result (Theorem 3).

5.4 The Complete Solution

Apply the quadratic formula (Theorem 3) to the resolvent:

$$w = t^3 = (-q \pm \sqrt{q^2 + 4p^3/27}) / 2$$

Take the real cube root of either branch (both yield the same u):

$$t = [(-q \pm \sqrt{q^2 + 4p^3/27}) / 2]^{1/3}$$

And recover the root:

$$u = t - p/(3t)$$

Theorem 6 (Cubic Formula from Fence). The real root of the depressed cubic $u^3 + pu + q = 0$ with $p > 0$ is:

$$u = t - p/(3t), \quad t = [(-q + \sqrt{q^2 + 4p^3/27})/2]^{1/3}$$

The formula requires one square root (from the degree-2 fence) and one cube root (from the absorption step). No Cardano $s + t$ decomposition is invoked.

Part III: The Derivation Chain

The complete logical chain, showing how each degree's result feeds the next:

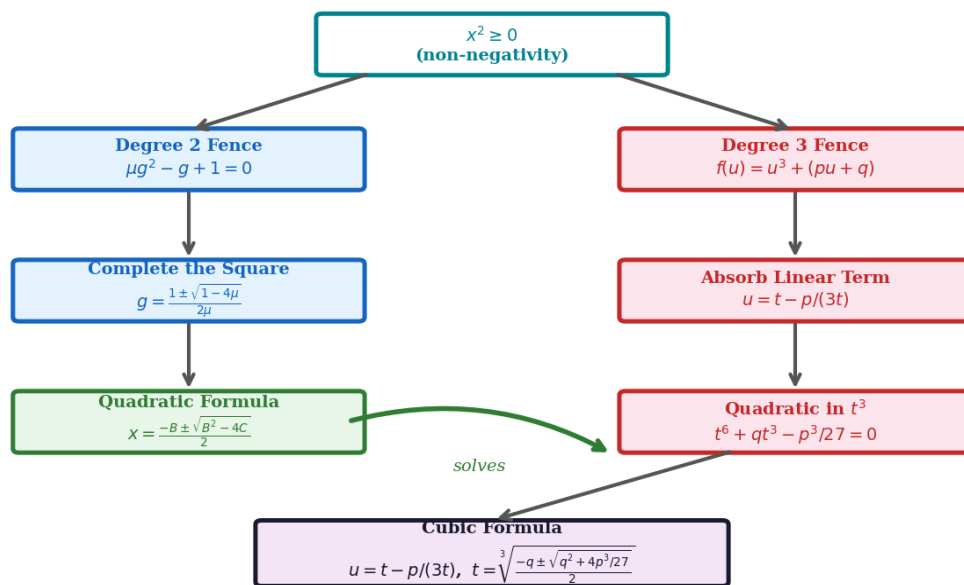


Figure 4. The derivation chain. Left column (blue): degree-2 path from fence to quadratic formula. Right column (red): degree-3 path from fence to resolvent quadratic. Green arrow: the quadratic formula solves the cubic's resolvent, completing the chain.

6. Comparison of Derivation Paths

Step	Cardano's Path (1545)	Fence Path (This Work)
Starting point	Guess $u = s + t$	Fence identifies inflection tangent
Key substitution	Impose $3st = -p$ (ad hoc)	Absorb linear term: $k = -p/3$ (forced)
Motivation	Algebraic cleverness	Calculus: eliminate t and $1/t$ terms
Resolvent	$s^3 + t^3, s^3t^3$ system	Quadratic in t^3 directly
Degree-2 solve	Generic quadratic formula	Degree-2 fence result (Theorem 3)
Self-contained?	Assumes quadratic formula	Derives quadratic formula from fence

The final formula is mathematically equivalent — it must be, since there is only one answer. What differs is the derivation: the fence path replaces Cardano's inspired guess ($u = s + t$ with $3st = -p$) with a calculus-motivated substitution ($u = t - p/(3t)$) that is uniquely forced by the requirement to absorb the inflection tangent's linear term into the cube. The entire chain is self-contained, deriving the quadratic formula from the degree-2 fence before using it to solve the cubic's resolvent.

Part IV: Worked Numerical Example

7. Example: $2x^3 + 5x^2 + 11x + 4 = 0$

Step 1: Depress the cubic. Divide by 2 and shift $x = u - 5/6$:

$$u^3 + (41/12)u - 77/54 = 0, \quad \text{so } p = 41/12 \approx 3.417, \quad q = -77/54 \approx -1.426.$$

Step 2: Verify one-real-root regime. $p = 41/12 > 0 \Rightarrow f'(u) = 3u^2 + 41/12 > 0$ for all u . Strictly monotone $\Rightarrow$ one real root. ✓

Step 3: Fence interval. $\alpha = 0$, $\beta = -q/p = (77/54)/(41/12) = 924/2214 \approx 0.4174$. Root lies in $[0, 0.4174]$.

Step 4: Apply the absorption substitution. $u = t - p/(3t) = t - 41/(36t)$.

Step 5: Form the resolvent quadratic. $w^2 + qw - p^3/27 = 0$:

$$w^2 - 1.42593w - 1.47726 = 0$$

Step 6: Solve with the quadratic formula (Theorem 3):

$$w = (-q \pm \sqrt{q^2 + 4p^3/27}) / 2 = (1.42593 \pm \sqrt{7.94213}) / 2$$

$$w_+ = 2.12205, \quad w_- = -0.69613$$

Step 7: Cube root. $t = w_+^{1/3} = (2.12205)^{1/3} = 1.28505$.

Step 8: Recover the root. $u = t - p/(3t) = 1.28505 - 0.88626 = 0.39878$.

Step 9: Unshift. $x = u - 5/6 = 0.39878 - 0.83333 = -0.43455$.

Verification: $2(-0.43455)^3 + 5(-0.43455)^2 + 11(-0.43455) + 4 = -0.000000 \approx 0$. ✓

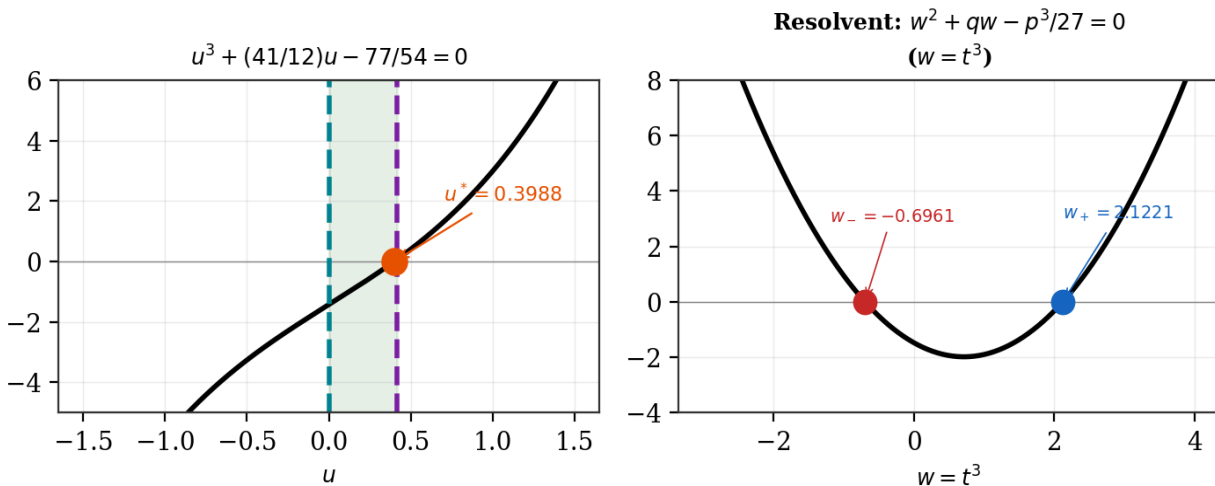


Figure 5. Left: the depressed cubic with fence interval (green shading) and root (orange). Right: the resolvent quadratic in $w = t^3$, with both roots marked. The positive root w_+ (blue) gives the real cube root.

8. Summary of the Method

Step	Operation	Result
1	Depress	$u^3 + pu + q = 0$ (shift by $-B/3$)
2	Check regime	$p > 0 \Rightarrow$ one real root ($f' > 0$ everywhere)
3	Fence interval	Root in $[\min(0, -q/p), \max(0, -q/p)]$
4	Identify structure	$f = u^3 +$ (inflection tangent)
5	Absorb	$u = t - p/(3t)$ eliminates linear term
6	Reduce	$t^3 + qt^3 - p^3/27 = 0$ (quadratic in t^3)
7	Solve quadratic	$t^3 = (-q \pm \sqrt{q^2 + 4p^3/27})/2$
8	Cube root	$t = (t^3)^{1/3}$

9	Root	$u = t - p/(3t)$
10	Unshift	$x = u - B/3$

Operations used: one $\sqrt{}$ (from degree-2 fence), one $\blacksquare$ (from absorption step), basic arithmetic. No Cardano $s + t$ decomposition. No guessing. The substitution $u = t - p/(3t)$ is uniquely forced by the absorption requirement.

Part V: Discussion

9. What Is New

The final formula is equivalent to Cardano's — it must be, since the cubic has a unique real root. What differs is the path:

(1) The entry point is the fence. The non-negativity constraint $x^2 \geq 0$ identifies the fence interval, which reveals the depressed cubic's structure as $u^3 +$ (inflection tangent).

(2) The substitution is forced, not guessed. Cardano's method requires the inspired guess $u = s + t$ and the ad hoc constraint $3st = -p$. In the fence path, the substitution $u = t - p/(3t)$ is the unique answer to the calculus question "what absorbs the linear term into the cube?" The value $k = -p/3$ is forced by requiring both the t and $1/t$ coefficients to vanish simultaneously.

(3) The chain is self-bootstrapping. The degree-2 fence derives the quadratic formula. The degree-3 fence produces a resolvent that is a quadratic in t^3 . The degree-2 result solves it. Each level feeds the next.

(4) The derivation is motivated throughout. Every step has a clear geometric or analytic justification: the fence identifies structure, calculus determines the substitution, and the degree-2 result closes the chain. There is no step that requires "trying something and seeing if it works."

10. Scope and Limitations

The derivation presented here covers the $p > 0$ (one-real-root) case. When $p < 0$ (potentially three real roots), the same substitution and reduction apply, but the quantity under the square root $q^2 + 4p^3/27$ may be negative (the casus irreducibilis). In that regime, the formula still works but requires cube roots of complex numbers — the same limitation as Cardano's formula. An alternative approach for $p < 0$, using the Fuss–Catalan series representation, is presented in Part II of this series.

11. Connection to Higher Degrees

The bootstrapping pattern suggests a general structure: the degree- d fence produces a universal equation whose resolvent is solvable by the degree- $(d-1)$ result. For degree 4, this would mean: degree-4 fence $\rightarrow$ absorption substitution $\rightarrow$ resolvent solvable by the cubic formula derived here. Whether this program can be carried out for the quartic (producing the quartic formula from the degree-4 fence and the degree-3 result) is an open question.

Appendix: The Role of the (a, b) Parameterization

A.1 The Exploration Tool

The derivation presented in Parts I–IV did not arrive fully formed. It was discovered through an exploration that used a different parameterization of the cubic's roots — one that does not appear in the final result but was essential in finding it. This appendix documents that exploration tool and its role in the discovery.

For the depressed cubic $u^3 + pu + q = 0$ with one real root, parameterize the roots as:

$$r_1 = -2a \text{ (real root)}, \quad r_2 = a + ci, \quad r_3 = a - ci$$

Vieta's relations give:

$$c^2 = p + 3a^2, \quad 2a(4a^2 + p) = q$$

The second relation is the *resolvent* $8a^3 + 2pa - q = 0$, a cubic in a . This showed that the entire problem reduces to finding one number: **a**. Once a is known, the real root is $r_1 = -2a$, and the complex pair follows from one square root: $r_{2,3} = a \pm i\sqrt{p + 3a^2}$.

A.2 How the Fence Brackets a

The fence applied to the resolvent gives $a \in [0, q/(2p)]$ when $p > 0$. This bracket, combined with the fixed-point iteration $a \rightarrow q/(2(4a^2 + p))$, produces a sequence of oscillating iterates that converge to a . The dimensionless reduction $a = (q/2p) \cdot g$ with $p = q^2/p^3$ yields the universal equation:

$$pg^3 + g = 1$$

This equation is a fixed point of the fence procedure — applying the fence to it reproduces itself. Power series expansion of $g(p)$ via Lagrange inversion gives:

$$g(p) = \sum [1/(3n+1)] C(3n+1, n) (-p)^n$$

where the coefficients $[1/(3n+1)]C(3n+1, n) = 1, 1, 3, 12, 55, 273, \dots$ are the **Fuss–Catalan numbers**, which count ternary trees with n internal nodes. This series converges when $|p| < 4/27$, and the convergence boundary $p = 4/27$ corresponds exactly to the vanishing of the discriminant $\Delta = -4p^3 - 27q^2 = 0$. The discriminant emerges from the convergence analysis, not assumed *a priori*.

A.3 The Search for a Closed Form

The Fuss–Catalan series is a genuine representation of the root, but it is infinite. The search for a closed-form (finite) expression drove the exploration through several intermediate attempts, each of which failed but narrowed the search:

Attempt	Idea	Outcome
1	Evaluate f at fence points, solve resulting system	Circular — both evaluations reduce to the original cubic
2	Parametric line intersecting cubic on fence interval	Self-similar — produces another cubic with same p
3	Parametric quadratic through both fence points	One free parameter; cubic factors through fences but root requires solving cubic for the parameter
4	Transform cubic into $(u+v)^3 + g = 0$	Key insight: at a root, $r^3 = -pr - q$ elimination gives a quadratic in r . Direct precursor to absorption.

Attempts 1–3 each hit the same wall: circularity. Attempt 4 was the breakthrough, and its full derivation reveals *why* the earlier approaches failed and how the absorption substitution escapes the trap.

A.3.1 Attempt 4 in Full: Reducing a Cubic to a Quadratic at a Root

We seek constants v and g such that at the root r of $u^3 + pu + q = 0$:

$$(r + v)^3 + g = 0$$

Expand $(r + v)^3$:

$$r^3 + 3vr^2 + 3v^2r + v^3 + g = 0$$

Since r is a root of the original cubic, we have the identity $r^3 = -pr - q$. Substitute:

$$(-pr - q) + 3vr^2 + 3v^2r + v^3 + g = 0$$

Collect by powers of r :

$$3vr^2 + (3v^2 - p)r + (v^3 + g - q) = 0$$

The r^3 term is gone. What remains is a **quadratic in r** . If we know v and g , the root is given by the quadratic formula:

$$r = [(p - 3v^2) \pm \sqrt{(3v^2 - p)^2 - 12v(v^3 + g - q)}] / (6v)$$

A cubic reduced to a square root — exactly the goal. But we have two unknowns (v and g) and only one constraint (the quadratic must yield a root of f). One free parameter remains undetermined, and pinning it down requires solving... a cubic. The circularity persists.

A.3.2 Why Working at a Root Is Circular

Attempts 1–4 all share a common structure: they begin with "let r be a root, then $r^3 = -pr - q$, therefore..." and attempt to extract information. But **any** equation derived by substituting $r^3 = -pr - q$ is merely a consequence of the cubic identity $r^3 + pr + q = 0$. We are rearranging a single equation, not introducing new information.

This is why every root-level approach failed:

Approach	Operates on	New information?
Evaluate f at fence points	The root r	No — rearranges $r^3 + pr + q = 0$
Parametric line through fence interval	The root r	No — self-similar, same p
Parametric quadratic through fence points	The root r	No — one free parameter remains cubic
$(r + v)^3 + g = 0$ at root	The root r	No — quadratic holds for any v
Vieta's relations	The roots collectively	No — equivalent to the cubic
$u = t - p/(3t)$	The polynomial itself	YES — new equation in t

The fundamental problem: *the cubic equation is the only algebraic relation the root satisfies*. Any manipulation that starts from " r is a root, therefore $r^3 = -pr - q$ " can only produce consequences of that one relation. No second independent equation can be obtained by rearranging the first.

A.3.3 How the Absorption Substitution Escapes

The substitution $u = t - p/(3t)$ does not work at a root. It does not begin with "let r be a root and substitute $r^3 = -pr - q$." Instead, it transforms the **entire polynomial as a function**:

$$f(u) = u^3 + pu + q \quad \blacksquare \quad h(t) = t^3 - p^3/(27t^3) + q$$

This is a **change of variable**, not an evaluation at a root. It operates on the polynomial itself, not on its solutions. The information comes from the *structure of the polynomial* — the algebraic relationship between the u^3 term and the pu term — not from the cubic identity at a root.

That is why it produces genuinely new information: $h(t) = 0$ multiplied by t^3 gives:

$$t^6 + qt^3 - p^3/27 = 0$$

This is a **different equation** — degree 6 in t , but quadratic in t^3 . It is not a rearrangement of the cubic. It is a new polynomial whose roots are related to the original roots by the substitution $u = t - p/(3t)$.

A.3.4 The Connection to the Fence

The fence told us that the depressed cubic decomposes as $f(u) = u^3 + (pu + q)$ — pure cube plus inflection tangent. This is a statement about **the function**, not about its roots. The absorption question — "what substitution makes $u^3 + pu$ a pure function of t^3 ?" — is also a question about **the function**, not about its roots.

This is the match. The fence provides structural information about the polynomial. The absorption uses structural information about the polynomial. Both operate at the level of the function, not at the level of solutions. That is why they connect, and that is why the result escapes the circularity that trapped every root-level approach.

In summary: working at a root traps you inside the cubic identity. The absorption substitution escapes by transforming the function *before* solving it.

A.4 Scaffolding vs. Destination

The (a, b) parameterization does not appear in the final derivation (Part II). The absorption substitution solves for u directly without ever isolating a . However, the parameterization was essential to the discovery in three ways:

(1) It showed the problem reduces to one unknown. The parameterization made visible that once a single number (a) is determined, everything else — the real root, the complex pair, the residual quadratic — follows from one square root. This gave the conviction that a closed form should exist.

(2) It generated the fence iteration and series. The resolvent $8a^3 + 2pa - q = 0$, living in the a -parameterization, produced the Fuss–Catalan series — a genuinely novel representation of the cubic root that stays real throughout (bypassing the casus irreducibilis) and connects polynomial root-finding to combinatorial tree enumeration.

(3) It guided the search through dead ends. Each failed attempt (parametric lines, parametric quadratics, fence-point evaluations) eliminated a class of approaches and narrowed the search until the absorption idea emerged.

A.5 Two Results from One Framework

The fence bound framework ultimately produced two distinct results for the cubic:

	Closed-Form Derivation (Part II)	Fuss–Catalan Series (This Appendix)
Uses (a, b) parameterization?	No — scaffolding only	Yes, centrally
Closed form?	Yes	No (infinite series)
Stays real (all cases)?	No (casus irreducibilis requires complex cube root)	Yes — all partial sums real
Novel?	New derivation of known formula	Genuinely new representation

Best for	One real root ($p > 0$)	Three real roots ($p < 0$)
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The first result (a new derivation of the known cubic formula) is the main contribution of this paper. The second (the Fuss–Catalan series) is a complementary result that lives entirely within the (a, b) parameterization and provides a purely real alternative for the three-real-root case. Together, they form a complete toolkit: the closed form handles the one-real-root case cleanly, and the series handles the three-real-root case without leaving the reals. Both originate from the same starting point: the fence bound $x^2 \geq 0$.

The (a, b) parameterization was the exploration tool. The absorption substitution was the destination. The tool was essential for finding the destination, even though it does not appear in the final formula. This is a common pattern in mathematical discovery: the scaffolding that supports construction is removed once the structure stands on its own.